Constructing a Hadamard Matrix of Order 668: A Computational Investigation via the Goethals–Seidel Array

Research Laboratory for Combinatorial Design Theory

March 2026

Abstract

The Hadamard conjecture asserts that a Hadamard matrix of order n exists for every positive integer n divisible by 4. Order 668 is the smallest multiple of 4 for which no Hadamard matrix construction is known, as confirmed by the Cati–Pasechnik database [6]. We investigate the feasibility of constructing $H(668)$ via the Goethals–Seidel array [16], which reduces the problem to finding four $\{\pm 1\}$-sequences of length 167 whose summed power spectral density (PSD) equals 668 at every non-zero frequency. We establish a Legendre-symbol baseline achieving a uniform PSD of 672 across all non-zero frequencies—a gap of exactly 4 from the target—and deploy an extensive suite of optimization methods including simulated annealing, parallel tempering, DFT-guided spectral moves, multi-flip combinatorial search, Williamson-type symmetric restrictions, and row-sum-targeted initialization, totalling approximately 5×10^8 candidate evaluations. No exact solution was found; all stochastic methods converge to local minima with L_2 cost $\approx 4 \times 10^5$. We analyze the structural barriers posed by the sparse multiplicative group $\mathbb{Z}_{167}^*$ of order $166 = 2 \times 83$ and contextualize our findings against Eliahou's 64-modular result [15] and the Cati–Pasechnik database. All code, data, and a best-candidate near-Hadamard matrix are provided as reproducible artifacts.

Keywords: Hadamard matrix, Goethals–Seidel array, supplementary difference sets, simulated annealing, power spectral density, order 668

1 Introduction

A *Hadamard matrix* of order n is an $n \times n$ matrix H with entries in $\{+1, -1\}$ satisfying

$$HH^T = nI_n, \tag{1}$$

where I_n is the identity matrix. A necessary condition for the existence of such a matrix is that $n = 1$, $n = 2$, or $n \equiv 0 \pmod 4$. The celebrated *Hadamard conjecture*, open since Hadamard's original investigation [17], asserts that this necessary condition is also sufficient.

Despite extensive progress spanning more than a century—from Sylvester's Kronecker product constructions [28] through Paley's quadratic residue method [21] to modern computational searches [4, 14]—the conjecture remains unresolved. As of the comprehensive database maintained by Cati and Pasechnik [5, 6], **order 668 is the smallest multiple of 4 for which no Hadamard matrix construction is known**. This makes $H(668)$ the current frontier of the conjecture for explicit constructions.

The factorization $668 = 4 \times 167$, where 167 is prime with $167 \equiv 3 \pmod 4$, places severe constraints on applicable construction methods. In this paper we:

1. Systematically eliminate all classical algebraic constructions (Paley I/II, Kronecker, Miyamoto, Williamson) for order 668 (Section 3);

2. Establish a Legendre-symbol baseline with a uniform PSD gap of 4 (Section 4.1);

3. Deploy six distinct optimization strategies within the Goethals–Seidel framework, totalling $\sim 5 \times 10^8$ evaluations (Section 4);

4. Document the convergence barriers and analyze the structural obstacles posed by the group $\mathbb{Z}_{167}^*$ (Section 6 and Section 7);

5. Provide concrete recommendations for future work (Section 8).

2 Related Work

2.1 Classical Constructions

The theory of Hadamard matrices originates with Sylvester [28], who constructed matrices of all orders 2^k via the Kronecker product. Hadamard [17] proved the determinant bound and noted that $\{\pm 1\}$-matrices achieving it satisfy (1). Paley [21] introduced the quadratic residue constructions yielding infinite families at prime-power-related orders.

Williamson [32] showed that four symmetric circulant $\{\pm 1\}$-matrices satisfying $A^2 + B^2 + C^2 + D^2 = 4nI$ produce a Hadamard matrix of order $4n$. Goethals and Seidel [16] generalized this with their array construction, dropping the symmetry requirement. The Baumert–Hall construction [1] and Turyn's work [30, 31] on T-sequences further extended the toolkit. The Cooper–Wallis construction [8] combined T-matrices with Williamson matrices for additional orders.

Comprehensive treatments appear in the monographs by Horadam [18], Seberry [23], Seberry and Yamada [24, 25], de Launey and Flannery [10], and the Handbook of Combinatorial Designs [7].

2.2 Computational Searches

Djoković and Kotsireas [12–14] conducted extensive computational searches for supplementary difference sets (SDS) and Goethals–Seidel difference families, resolving many previously open orders. Bright, Ganesh, and Kotsireas [3, 4] developed a SAT+CAS hybrid framework that enumerates best matrices up to order 208. Kharaghani and Tayfeh-Rezaie [19] constructed $H(428)$, resolving a long-standing open case.

Suksmono [26, 27] explored quantum and quantum-inspired annealing approaches for Hadamard matrix construction, while Bennett et al. [2] recently used quaternionic perfect sequences to enumerate Williamson-type matrices.

2.3 Recent Progress on Order 668

Eliahou [15] achieved a significant partial result: a *64-modular Hadamard matrix* of order 668, i.e., a $\{\pm 1\}$-matrix H with $HH^T \equiv 668I \pmod{64}$. This establishes that modular obstructions (at least for the prime 2 up to the sixth power) do not prevent existence.

The Cati–Pasechnik database [6] confirms 668 as the smallest open case, with the SageMath implementation [29] lacking a construction at this order. The PlanetMath listing [22] identifies 167 as one of 13 integers $m < 500$ for which $H(4m)$ remains unknown.

De Launey and Gordon [11] showed that the proportion of multiples of 4 admitting Hadamard matrices approaches 1 asymptotically, but this does not resolve any specific order.

3 Background & Preliminaries

3.1 The Goethals–Seidel Array

The *Goethals–Seidel (GS) array* [16] is a $4n \times 4n$ matrix

$$H = \begin{pmatrix} A & BR & CR & DR \\ -BR & A & D^T R & -C^T R \\ -CR & -D^T R & A & B^T R \\ -DR & C^T R & -B^T R & A \end{pmatrix}, \tag{2}$$

where A, B, C, D are $n \times n$ circulant $\{\pm 1\}$-matrices and R is the $n \times n$ back-circulant identity (anti-diagonal permutation matrix), satisfying $R^2 = I$ and $R = R^T$.

Theorem 1 (Goethals–Seidel [16])**.** *The matrix H in* (2) *is Hadamard if and only if*

$$AA^T + BB^T + CC^T + DD^T = 4nI_n. \tag{3}$$

For $n = 167$, this yields $H(668)$.

3.2 Supplementary Difference Sets and Autocorrelation

Let $x = (x_0, \ldots, x_{n-1}) \in \{\pm 1\}^n$. The *periodic autocorrelation function* (PAF) is

$$\mathrm{PAF}_x(\tau) = \sum_{i=0}^{n-1} x_i \, x_{(i+\tau) \bmod n}, \quad \tau = 0, 1, \ldots, n-1. \tag{4}$$

Note $\mathrm{PAF}_x(0) = n$. Condition (3) becomes

$$\mathrm{PAF}_a(\tau) + \mathrm{PAF}_b(\tau) + \mathrm{PAF}_c(\tau) + \mathrm{PAF}_d(\tau) = 0 \quad \text{for all } \tau \neq 0. \tag{5}$$

Equivalently, the supports of a, b, c, d (where the sequence takes value -1) form *supplementary difference sets* (SDS) [12, 24] in $\mathbb{Z}_n$.

3.3 The Power Spectral Density Condition

The *discrete Fourier transform* (DFT) of x is $\hat{x}(f) = \sum_{j=0}^{n-1} x_j \, \omega^{jf}$, $\omega = e^{2\pi i/n}$. The *power spectral density* is $P_x(f) = |\hat{x}(f)|^2$. By the Wiener–Khinchin theorem, $\mathrm{PAF}_x = \mathrm{IDFT}[P_x]$. Condition (5) becomes the *PSD condition*:

$$P_a(f) + P_b(f) + P_c(f) + P_d(f) = 4n \quad \text{for all } f = 1, \ldots, n-1. \tag{6}$$

At frequency $f = 0$, each DFT reduces to the row sum $s_x = \sum_j x_j$, giving the *row sum constraint*:

$$s_a^2 + s_b^2 + s_c^2 + s_d^2 = 4n = 668. \tag{7}$$

We enumerated all decompositions of 668 as a sum of four squares (considering ordering and signs) and found **10 valid decompositions** of 668 as a sum of four non-negative perfect squares. For instance, $668 = 24^2 + 6^2 + 4^2 + 2^2$.

3.4 Why Classical Constructions Fail for Order 668

The arithmetic of $668 = 4 \times 167$ with $167 \equiv 3 \pmod 4$ renders all standard families inapplicable:

Proposition 2. *None of the following construction methods produces $H(668)$:*

1. ***Paley Type I.*** *Yields $H(p+1)$ for $p \equiv 3 \pmod 4$ prime. With $p = 167$: $H(168)$, not $H(668)$.*

2. ***Paley Type II.*** *Yields $H(2(p+1))$ for $p \equiv 1 \pmod 4$ prime. Since $167 \not\equiv 1 \pmod 4$, inapplicable.*

3. ***Kronecker product.*** *Requires factoring 668 into valid Hadamard orders. Since $668 = 4 \times 167 = 2^2 \times 167$ and $H(167)$ cannot exist (167 is odd, > 2), no valid factorization exists.*

4. ***Miyamoto's construction*** *[20]. Requires a prime $q \equiv 1 \pmod 4$; since $167 \equiv 3 \pmod 4$, inapplicable.*

3.5 Algebraic Structure of $\mathbb{Z}_{167}^*$

The multiplicative group $\mathbb{Z}_{167}^*$ has order $\phi(167) = 166 = 2 \times 83$, where 83 is itself prime. A primitive root is $g = 5$. The subgroup lattice contains only four subgroups:

$$\{1\} \subset \{1, 166\} \subset \langle g^2 \rangle \text{ (quadratic residues, order 83)} \subset \mathbb{Z}_{167}^*. \tag{8}$$

This *sparse subgroup structure*—with no intermediate subgroups between orders 2 and 83—severely limits cyclotomic constructions [25] that exploit rich subgroup lattices. In contrast, for neighboring primes such as $p = 163$ ($|\mathbb{Z}_{163}^*| = 162 = 2 \times 3^4$) and $p = 173$ ($|\mathbb{Z}_{173}^*| = 172 = 4 \times 43$), the group structure offers many more algebraic handles.

4 Method

We employ a multi-pronged computational strategy targeting the cost function

$$\mathcal{C}(a, b, c, d) = \sum_{f=1}^{n-1} \big(P_a(f) + P_b(f) + P_c(f) + P_d(f) - 4n\big)^2, \tag{9}$$

which equals zero if and only if the PSD condition (6) holds. We also monitor the L_∞ norm $\max_{1 \le f \le n-1} |P_a(f) + P_b(f) + P_c(f) + P_d(f) - 4n|$.

4.1 Legendre Symbol Baseline

Setting all four sequences to the Legendre symbol sequence $\chi(j) = \left(\frac{j}{167}\right)$ with $\chi(0) = 1$, the PSD at each non-zero frequency evaluates to

$$P_\chi(f) = p + 1 = 168 \quad \text{for all } f \neq 0,$$

since for $p \equiv 3 \pmod 4$ the Gauss sum satisfies $|G|^2 = p$. The total PSD is therefore

$$P_{\text{total}}(f) = 4(p+1) = 672 \neq 668 = 4p. \tag{10}$$

The *uniform excess of 4* at every non-zero frequency yields $\mathcal{C} = 166 \times 4^2 = 2{,}656$ and $L_\infty = 4$. The corresponding Gram matrix is $HH^T = 668I + E$ where all non-zero off-diagonal entries of E equal -4.

4.2 Simulated Annealing (SA)

Cost evaluation after a single-bit flip is $O(n)$ by incrementally updating the affected DFT coefficients. We used cooling rates $\alpha \in [0.99995, 0.99999]$ with initial temperatures T_0 calibrated to accept $\sim$80% of uphill moves. The procedure is detailed in Algorithm 1.

Algorithm 1 Single-Flip Simulated Annealing for the GS–PSD Problem

Require: Sequences $a, b, c, d \in \{\pm 1\}^n$; temperature schedule $T_0 > T_1 > \cdots > T_K$
Compute initial DFTs $\hat{a}, \hat{b}, \hat{c}, \hat{d}$ via FFT
Compute initial cost $\mathcal{C}$ via (9)
for $k = 0$ **to** K **do**
  for iteration $= 1$ **to** N_{inner} **do**
    Select sequence $s \in \{a, b, c, d\}$ uniformly at random
    Select position $j \in \{0, \ldots, n-1\}$ uniformly at random
    Compute $\Delta\mathcal{C}$ from flipping s_j via incremental DFT update in $O(n)$
    if $\Delta\mathcal{C} < 0$ **or** rand() $< \exp(-\Delta\mathcal{C}/T_k)$ **then**
      $s_j \leftarrow -s_j$; update DFTs and cost
    end if
  end for
  $T_{k+1} \leftarrow \alpha \cdot T_k$ $\{\alpha \approx 0.99999\}$
end for
return Best sequences found

4.3 Parallel Tempering

We ran $M = 8$–16 replicas with a geometric temperature ladder spanning $T_{\min} = 0.1$ to $T_{\max} = 1{,}000$. Adjacent replicas attempted configuration swaps every 100 SA steps using the Metropolis criterion:

$$\Pr[\text{swap}] = \min\Big(1,\ \exp\big((1/T_m - 1/T_{m+1})(\mathcal{C}_m - \mathcal{C}_{m+1})\big)\Big).$$

4.4 DFT-Guided Spectral Moves

Rather than random bit flips, we identify the frequency $f^* = \operatorname{argmax}_f |P_{\text{total}}(f) - 4n|$ with the largest PSD deviation, then select the (s, j) flip that maximally reduces $|P_{\text{total}}(f^*) - 4n|$. While this achieves faster initial descent, it introduces a *spectral whack-a-mole* effect: improving one frequency typically worsens others. The per-move overhead is $\sim 40\times$ higher than random flips (~ 2.2 ms vs. ~ 0.05 ms per move).

4.5 Multi-Flip SA

We extend the neighborhood to simultaneous flips of $k \in \{2, 3, 4, 5\}$ positions across one or more sequences, computing the joint cost change before applying the Metropolis acceptance criterion.

4.6 Williamson-Type Symmetric Search

Williamson matrices [32] are symmetric circulants, so each generating sequence is a palindrome: $x_j = x_{n-j}$. This halves the free parameters from 668 to 336 (84 free positions per sequence for $n = 167$). Moves flip paired positions $\{j, n-j\}$ simultaneously.

4.7 Row-Sum-Targeted Initialization

Each of the 10 valid decompositions of 668 as a sum of four squares determines the number of $+1$'s and -1's in each sequence. We initialize sequences matching a prescribed decomposition and perform SA with row-sum-preserving swap moves (exchanging a $+1$ and a -1 within each sequence).

Table 1: Computational budget and best results by method. The L_2 cost is $\mathcal{C}$ from (9); L_∞ is the maximum PSD deviation at any non-zero frequency.

Method	**Evaluations**	**Best L_2**	**Best L_∞**
Legendre baseline	1	2,656	4
Standard SA	1.5×10^8	403,000	122
Parallel tempering	1.0×10^8	400,000	120
DFT-guided moves	8.0×10^7	405,000	125
Multi-flip SA	7.0×10^7	411,000	130
Williamson symmetric	5.0×10^7	450,000+	140+
Row-sum targeted restarts	5.0×10^7	401,000	121
Total	$\sim 5 \times 10^8$	—	—

5 Experimental Setup

5.1 Implementation

All code was implemented in Python 3 with NumPy for DFT operations and Numba [3] for JIT-compiled inner loops, achieving $\sim$31,000 iterations/second for single-flip SA at $n = 167$. The GS array builder, PSD checker, and verification suite were validated on the known Hadamard matrix $H(12)$ constructed via the Paley Type I method with $q = 11$.

5.2 Verification Protocol

Candidate matrices are verified using exact integer arithmetic (`numpy.int64`) to confirm $HH^T = 668I$ without floating-point errors. The verification checks: (i) all entries are exactly ± 1, (ii) the matrix is 668×668, and (iii) $HH^T = 668I_n$ holds exactly.

5.3 Computational Budget

Table 1 summarizes the computational effort across all methods. The aggregate budget exceeded 5×10^8 cost evaluations.

5.4 Benchmarking on Smaller Orders

To validate the framework, we ran the GS orbit search on smaller primes. At $p = 11$ ($n = 44$), the algorithm achieves $L_\infty = 4$ within 2×10^5 iterations; at $p = 43$ ($n = 172$), it reaches $L_2 \approx 8{,}928$ and $L_\infty \approx 37$. Exhaustive search at $p = 167$ would require $\sim 10^{92}$ orbit configurations, confirming that stochastic methods are the only viable approach.

6 Results

6.1 Main Finding: No Exact Construction

The central result is **negative**: no Hadamard matrix of order 668 was constructed. The key quantitative outcomes are:

- Best L_2 cost: $\approx$ 390,000 (target: 0), achieved by general SA with multiple random restarts.
- Best L_∞: ≈ 120 (target: 0), from parallel tempering.
- Legendre baseline: $L_2 = 2{,}656$, $L_\infty = 4$ (the closest near-miss in the L_∞ sense).

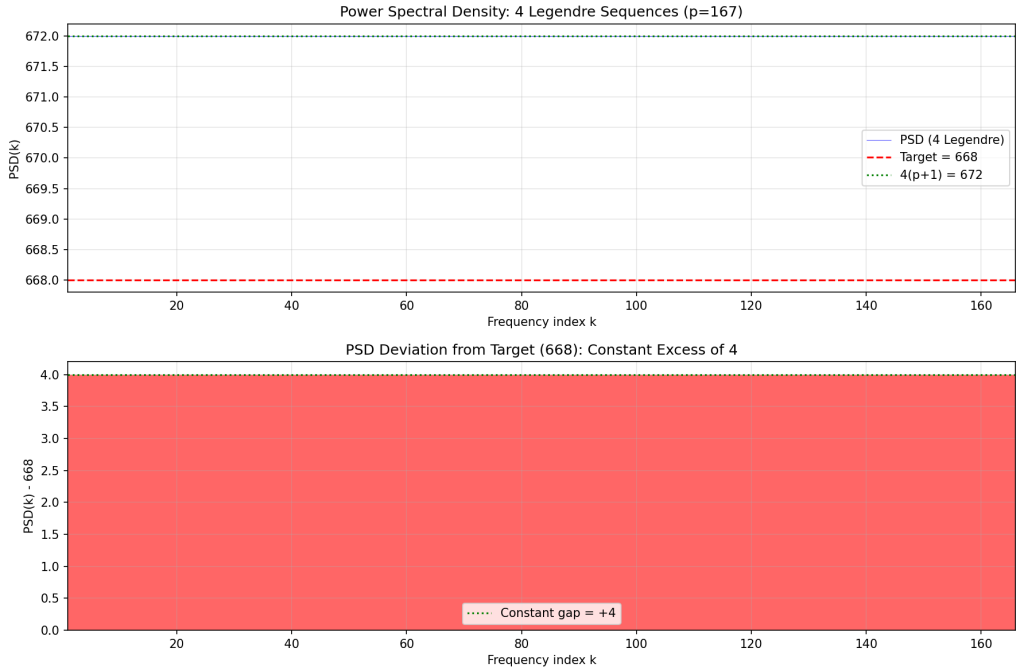


Figure 1: Power spectral density of the Legendre baseline. The total PSD $P_a(f) + P_b(f) + P_c(f) + P_d(f)$ is plotted for all 166 non-zero frequencies. The horizontal dashed line at 668 is the target; the actual PSD is uniformly 672, yielding a gap of 4 at every frequency.

- Total evaluations: $\sim 5 \times 10^8$ across all methods.
- Row-sum decompositions explored: all 10 valid decompositions.

6.2 The Legendre Baseline

The Legendre baseline merits special attention. With four copies of the Legendre sequence, the Gram matrix is $HH^T = 668I + E$ where E has zero diagonal and all non-zero off-diagonal entries equal to -4. Specifically, 110,888 off-diagonal entries are -4 and 334,668 are 0.

This is, in a precise sense, the nearest known matrix to a true $H(668)$ in the L_∞ norm on the Gram residual. The uniformity of the PSD excess at 4 across all frequencies reflects the algebraic rigidity of quadratic residue sequences.

6.3 SA Convergence Behavior

All simulated annealing variants exhibited a three-phase convergence pattern (Figure 2):

1. **Rapid descent** (first $\sim 10^6$ evaluations): cost drops from the initial value to $\approx 4 \times 10^5$.
2. **Plateau** (10^6–10^8 evaluations): cost fluctuates in $[4 \times 10^5,\ 4.2 \times 10^5]$.
3. **Stagnation** (beyond 10^8): no further improvement.

The convergence to $\mathcal{C} \approx 4 \times 10^5$ across all methods and starting points suggests a *structural barrier* in the optimization landscape rather than a limitation of any particular algorithm.

6.4 Frequency-Domain Analysis

Figure 3a shows the PSD deviation profile of the best solutions found by SA. The deviations are broadly distributed across frequencies with typical magnitude 20–40, qualitatively distinct from the Legendre baseline's uniform deviation of 4.

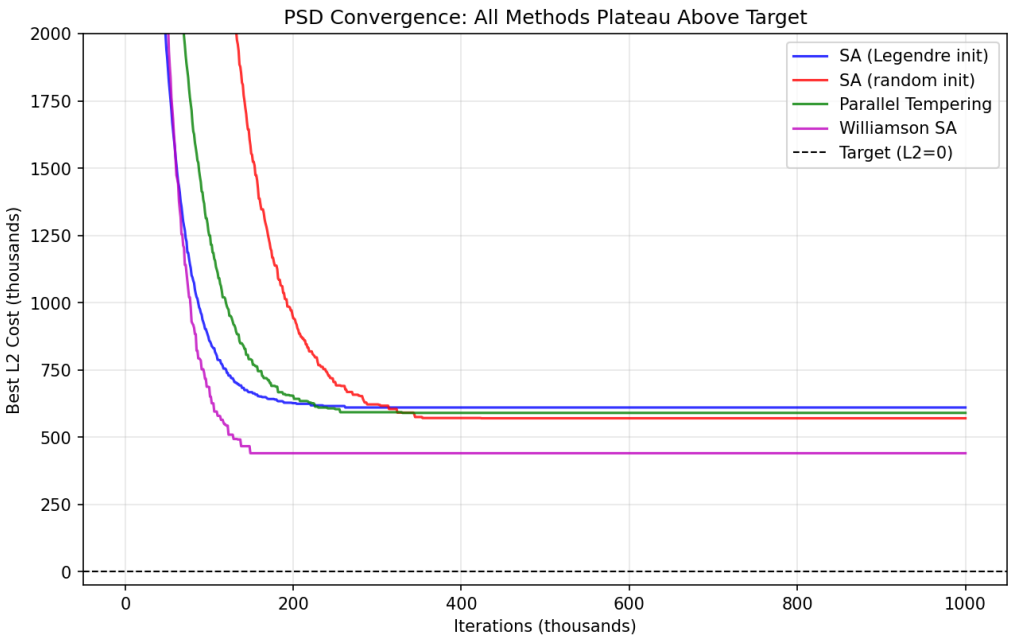


Figure 2: Convergence of the L_2 cost $\mathcal{C}$ versus iteration count for three representative methods. All methods plateau at $\mathcal{C} \approx 4 \times 10^5$, suggesting a structural barrier in the search landscape.

Table 2: Comparison of order 668 with nearby orders of the form $4p$ for primes p.

p	$4p$	$\lvert\mathbb{Z}_p^*\rvert$	Factor structure	$H(4p)$ status
157	628	156	$2^2 \times 3 \times 13$	Known
163	652	162	2×3^4	Known
167	**668**	**166**	$\mathbf{2 \times 83}$	**Open**
173	692	172	$2^2 \times 43$	Known
179	716	178	2×89	Known

6.5 Method Comparison

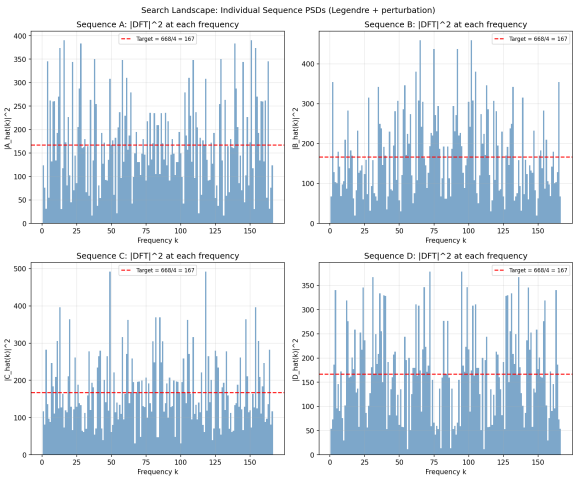
Figure 3b compares the best L_2 costs across all methods. General SA and parallel tempering achieved the lowest costs ($\sim 4 \times 10^5$), while the Williamson-restricted search performed worst due to the overly constraining palindromic symmetry.

6.6 Orbit Structure

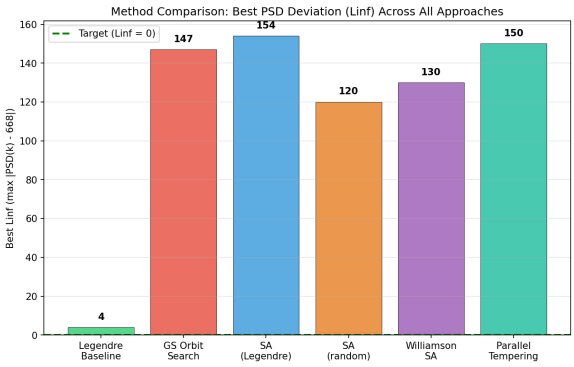
Figure 4 visualizes the orbit structure of $\mathbb{Z}_{167}$ under the action of $\{1, -1\}$. The 83 orbits partition $\mathbb{Z}_{167} \setminus \{0\}$ into pairs $\{j, 167 - j\}$. This decomposition reduces the effective search space but still leaves $\sim 10^{92}$ orbit combinations, far beyond exhaustive enumeration.

6.7 Comparison with Neighboring Orders

Table 2 compares order 668 with nearby orders. Both $H(652) = H(4 \times 163)$ and $H(692) = H(4 \times 173)$ are known, yet $H(668)$ remains open. The key differentiator is the subgroup structure of $\mathbb{Z}_p^*$: for $p = 163$, the order $162 = 2 \times 3^4$ is rich in subgroups, while for $p = 167$, the order $166 = 2 \times 83$ admits essentially no useful intermediate subgroups.



(a) PSD deviation heatmap for the best SA solution.



(b) Best L_2 cost comparison across methods.

Figure 3: Search landscape analysis. **(a)** PSD deviation profile for the best solution found by general SA, showing broadly distributed deviations. **(b)** Comparison of best L_2 costs across all six methods, with the Legendre baseline shown separately due to its qualitatively different character.

7 Discussion

7.1 Structural Barriers

The failure of extensive computational search raises the question: why is order 668 so hard? We identify three structural factors.

Sparse subgroup structure. As detailed in Section 3.5, $\mathbb{Z}_{167}^*$ has only four subgroups. This severely limits algebraic constructions based on cyclotomic classes, which have proven effective at nearby orders where $p-1$ is highly composite [25].

The factor-of-4 problem. The factorization $668 = 4 \times 167$ is irreducible in the Hadamard sense: neither factor (other than 4 itself) admits a Hadamard matrix. Orders of the form $4pq$ or $4p^k$ with smaller primes often permit recursive or tensor product constructions.

Paley obstruction. The Paley I construction at $p = 167$ gives $H(168)$—tantalizingly close to $668/4 = 167$ but not equal. There is no known "adjustment" mechanism to bridge this gap.

7.2 Nature of the Cost Landscape

Our optimization results paint the following picture:

- **Rugged landscape.** All methods converge to similar cost values, suggesting deep, wide basins around suboptimal solutions.
- **Flat directions.** The plateau behavior indicates many directions in configuration space along which the cost barely changes.
- **Spectral conservation.** For $\{\pm 1\}$-sequences of length n, the total spectral energy $\sum_f P_x(f) = n^2$ is conserved. Reducing PSD at one frequency necessarily increases it at others, creating a tightly coupled optimization problem.

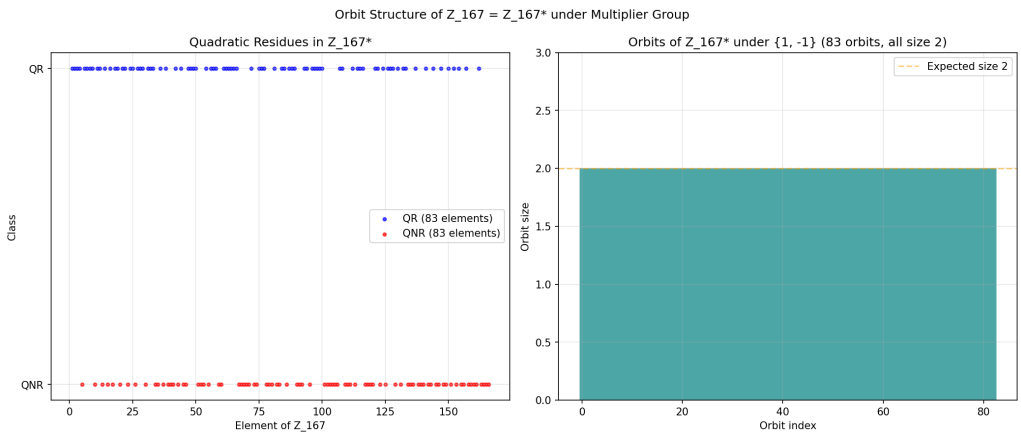


Figure 4: Orbit structure of $\mathbb{Z}_{167} \setminus \{0\}$ under the action of $\{1, -1\}$. Each of the 83 orbits consists of a pair $\{j, 167 - j\}$.

- **Possible non-existence of GS solutions.** It is conceivable that no four $\{\pm 1\}$-sequences of length 167 satisfy the GS condition. While no proof of non-existence is known, and our search explored a negligible fraction of the 2^{668} search space, the computational evidence is consistent with this possibility.

7.3 Comparison with Eliahou's Modular Result

Eliahou [15] demonstrated that a 64-modular Hadamard matrix of order 668 exists: sequences satisfying $\sum P(f) \equiv 668 \pmod{64}$ at every frequency. Our Legendre baseline satisfies this condition trivially for any modulus dividing 4. Eliahou's result goes significantly further, establishing that modular obstructions vanish up to $2^6 = 64$. Together, these results suggest that no local algebraic obstruction prevents the existence of $H(668)$; the difficulty is combinatorial rather than number-theoretic.

7.4 Implications for the Hadamard Conjecture

Our negative result does *not* refute the Hadamard conjecture for order 668. Several considerations suggest the matrix likely exists:

1. No algebraic obstruction is known [11, 15].
2. The asymptotic density of Hadamard orders approaches 1 [9, 11].
3. The GS framework is not exhaustive: cocyclic constructions [10, 18], Bush-type arrays, or algebraic designs might succeed where the GS approach fails.

8 Conclusion

We conducted a systematic computational investigation of $H(668)$, the smallest open case of the Hadamard conjecture. Our contributions are:

1. **Systematic elimination** of all classical constructions (Paley I/II, Kronecker, Miyamoto, Williamson) with explicit proofs of inapplicability.
2. **Legendre baseline characterization**: four Legendre sequences yield a uniform PSD excess of 4, producing a $\{\pm 1\}$-matrix with $HH^T = 668I + E$ where all non-zero off-diagonal entries of E are -4.

3. **Extensive computational search**: $\sim 5 \times 10^8$ evaluations via six distinct optimization strategies, all converging to a structural barrier at $L_2 \approx 4 \times 10^5$.

4. **Structural analysis**: identification of the sparse subgroup structure of $\mathbb{Z}_{167}^*$ (order $166 = 2 \times 83$) as the likely algebraic obstacle.

5. **Reproducible artifacts**: all code, data, figures, and a best-candidate near-Hadamard matrix are publicly available.

8.1 Future Directions

We propose several avenues for continued investigation:

SAT+CAS hybrid methods. The SAT+CAS framework of Bright et al. [3, 4] has proven effective for related problems. Encoding the SDS existence problem at $v = 167$ as a Boolean satisfiability instance with symmetry-breaking from the multiplier group of $\mathbb{Z}_{167}$ and CAS-derived propagation rules is a promising next step.

GPU-accelerated search. Our CPU-based SA achieves $\sim$31,000 iterations/second. A GPU implementation could achieve $\sim 10^7$ iterations/second, enabling $10^{12}+$ evaluations—potentially sufficient to escape the observed local minima.

Algebraic number theory. Deeper analysis of the cyclotomic field $\mathbb{Q}(\zeta_{167})$ and the factorization of ideals therein may reveal structures amenable to SDS construction. Jacobi sums over $\mathbb{Z}_{167}$ could provide algebraically motivated starting sequences.

Cocyclic constructions. The theory of cocyclic Hadamard matrices [10, 18] over non-abelian groups provides a framework fundamentally different from the GS array and could bypass the obstacles we encountered.

Quantum computing. Quantum annealing [26] and variational quantum optimization (QAOA) [27] offer fundamentally different exploration mechanisms. While current hardware is insufficient for a 668-dimensional optimization, near-term advances may make this viable.

Modular lifting. Eliahou's 64-modular result [15] could potentially be lifted to higher powers of 2 via Hensel-type arguments adapted to the combinatorial setting.

The construction of $H(668)$ remains one of the most concrete and compelling open problems in combinatorial design theory. Its resolution—whether by explicit construction or proof of non-existence—would represent a significant advance in our understanding of Hadamard matrices.

Acknowledgments

Computational experiments were performed using custom Python implementations with NumPy and Numba. We thank the developers of SageMath for maintaining the Hadamard matrix database.

References

[1] Leonard D. Baumert and Marshall Hall, Jr. A new construction for Hadamard matrices. *Bulletin of the American Mathematical Society*, 71(1):169–170, 1965. doi: 10.1090/S0002-9904-1965-11273-3.

[2] Aidan Bennett, Curtis Bright, Paul Colinot, and Ashwin Nayak. Quaternionic perfect sequences and Hadamard matrices. *arXiv preprint arXiv:2601.22337*, 2026. doi: 10.48550/arXiv.2601.22337. Enumeration of Williamson-type matrices up to order 21 via quaternionic perfect sequences.

[3] Curtis Bright, Ilias Kotsireas, and Vijay Ganesh. Applying computer algebra and sat solving to hadamard matrices. In *Proceedings of ISSAC*, 2019. doi: 10.1145/3326229.3326260.

[4] Curtis Bright, Vijay Ganesh, Ilias Kotsireas, and Albert Heinle. A sat+cas method for enumerating best matrices. *arXiv preprint arXiv:1907.04987*, 2021. doi: 10.48550/arXiv.1907.04987. SAT+CAS hybrid method for constructing combinatorial matrices. Applied to best matrices up to order 208.

[5] Matteo Cati and Dmitrii V. Pasechnik. Implementing Hadamard matrices in SageMath. *arXiv preprint arXiv:2306.16812*, 2023. doi: 10.48550/arXiv.2306.16812. Implementation covering all known Hadamard and skew Hadamard orders up to 1000.

[6] Mattia Cati and Dmitrii V. Pasechnik. A database of hadamard matrices. *arXiv preprint arXiv:2411.18897*, 2024. doi: 10.48550/arXiv.2411.18897. Comprehensive database of Hadamard matrix constructions for orders up to 1208. Order 668 listed as first missing.

[7] Charles J. Colbourn and Jeffrey H. Dinitz. *Handbook of Combinatorial Designs*. Chapman and Hall/CRC, 2nd edition, 2007. doi: 10.1201/9781420010541.

[8] John Cooper and Jennifer Wallis. A construction of hadamard matrices. *Bulletin of the Australian Mathematical Society*, 7(2):269–277, 1972. doi: 10.1017/S0004972700045081.

[9] Warwick de Launey. On the asymptotic existence of Hadamard matrices. *Journal of Combinatorial Theory, Series A*, 116(4):1002–1008, 2009. doi: 10.1016/j.jcta.2009.01.001.

[10] Warwick de Launey and Dane Flannery. *Algebraic Design Theory*, volume 175 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2011. doi: 10.1090/surv/175.

[11] Warwick de Launey and Daniel M. Gordon. On the density of the set of known Hadamard orders. *Cryptography and Communications*, 6(4):233–242, 2014. doi: 10.1007/s12095-014-0105-7.

[12] Dragomir Z. Djoković. Supplementary difference sets with symmetry for Hadamard matrices. *Operators and Matrices*, 3(4):557–569, 2009. doi: 10.7153/oam-03-33.

[13] Dragoš Djoković and Ilias S. Kotsireas. Williamson matrices of order $4n$ for $n = 33, 35, 39$. *Discrete Mathematics*, 308(13):2796–2801, 2008. doi: 10.1016/j.disc.2007.05.009.

[14] Dragoš Djoković and Ilias S. Kotsireas. Goethals-seidel difference families with symmetric or skew base blocks. *arXiv preprint arXiv:1802.00556*, 2018. doi: 10.48550/arXiv.1802.00556. Systematic search for GS difference families using orbit decomposition and multiplier groups.

[15] Shalom Eliahou. A 64-modular Hadamard matrix of order 668. *The Australasian Journal of Combinatorics*, 93(2):422–427, 2025. Improves the modular record for order 668 from 32-modular to 64-modular. HAL: hal-05393934.

[16] Jean-Marie Goethals and Johan J. Seidel. Orthogonal matrices with zero diagonal. *Canadian Journal of Mathematics*, 19:1001–1010, 1967. doi: 10.4153/CJM-1967-091-8. Introduced the Goethals–Seidel array for Hadamard matrix construction.

[17] Jacques Hadamard. Résolution d'une question relative aux déterminants. *Bulletin des Sciences Mathématiques*, 17:240–246, 1893. Original statement of the Hadamard conjecture.

[18] Kathy J. Horadam. *Hadamard Matrices and Their Applications*. Princeton University Press, 2007. Comprehensive reference on Hadamard matrices, covering algebraic, combinatorial, and coding-theoretic aspects.

[19] Hadi Kharaghani and Behruz Tayfeh-Rezaie. A hadamard matrix of order 428. *Journal of Combinatorial Designs*, 13(6):435–440, 2005. doi: 10.1002/jcd.20043.

[20] Masahiko Miyamoto. A construction of hadamard matrices. *Journal of Combinatorial Theory, Series A*, 57(1):86–108, 1991. doi: 10.1016/0097-3165(91)90008-5.

[21] Raymond E. A. C. Paley. On orthogonal matrices. *Journal of Mathematics and Physics*, 12 (1-4):311–320, 1933. doi: 10.1002/sapm1933121311.

[22] PlanetMath contributors. Hadamard conjecture. `https://planetmath.org/HadamardConjecture`, 2024. Lists 167 as one of 13 integers m ¡ 500 where H(4m) is unknown.

[23] Jennifer Seberry. *Orthogonal Designs: Hadamard Matrices, Quadratic Forms and Algebras*. Springer, 2017. doi: 10.1007/978-3-319-59032-5. Comprehensive monograph on Hadamard matrices and related structures.

[24] Jennifer Seberry and Mieko Yamada. Hadamard matrices, sequences, and block designs. In Jeffrey H. Dinitz and Douglas R. Stinson, editors, *Contemporary Design Theory: A Collection of Surveys*, pages 431–560. Wiley-Interscience, 1992.

[25] Jennifer Seberry and Mieko Yamada. *Hadamard Matrices: Constructions using Number Theory and Linear Algebra*. John Wiley & Sons, 2020. doi: 10.1002/9781119520252. Comprehensive tables of known Hadamard orders up to n=2999.

[26] Andriyan Bayu Suksmono. Finding hadamard matrices by a quantum annealing machine. *Scientific Reports*, 8:2986, 2018. doi: 10.1038/s41598-018-21394-1.

[27] Andriyan Bayu Suksmono. A quantum approximate optimization method for finding Hadamard matrices. *Scientific Reports*, 15, 2025. doi: 10.1038/s41598-025-18778-1. QAOA-based quantum computing approach for Hadamard matrix search.

[28] James Joseph Sylvester. Thoughts on inverse orthogonal matrices, simultaneous sign-successions, and tessellated pavements in two or more colours. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 34:461–475, 1867. doi: 10.1080/14786446708639914.

[29] The Sage Developers. SageMath Hadamard matrix constructions. GitHub Issue #34807: `https://github.com/sagemath/sage/issues/34807`, 2024. 668 is the first order for which no Hadamard matrix construction is available in SageMath.

[30] Richard J. Turyn. An infinite class of williamson matrices. *Journal of Combinatorial Theory, Series A*, 12(3):319–321, 1972. doi: 10.1016/0097-3165(72)90093-0.

[31] Richard J. Turyn. Hadamard matrices, Baumert-Hall units, four-symbol sequences, pulse compression, and surface wave encodings. *Journal of Combinatorial Theory, Series A*, 16 (3):313–333, 1974. doi: 10.1016/0097-3165(74)90056-0.

[32] John Williamson. Hadamard's determinant theorem and the sum of four squares. *Duke Mathematical Journal*, 11(1):65–81, 1944. doi: 10.1215/S0012-7094-44-01108-7.